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Blood Pressure–Lowering Effects of Aldosterone Synthase Inhibitors—A Systematic Review

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ABSTRACT

Excess aldosterone production contributes to the development of hypertension and results in fibrosis with dysfunction of the heart, vasculature and kidneys. Consequently, new agents have been developed to reduce endogenous aldosterone synthesis. The primary objective of this systematic review is to describe the BP-lowering effects of aldosterone synthase inhibitors (ASIs) in hypertensive patients and, secondly, to describe their potential renal protective effects and possible influence on cortisol production and plasma potassium. We searched PubMed, Embase and [ClinicalTrials.gov](https://www.clinicaltrials.gov) and included randomized controlled and clinical trials according to PICO using the review tool Covidence. Thirteen studies were included and all demonstrated BP reduction through ASI treatment. Among patients with apparent resistant hypertension, the placebo-corrected reductions in seated systolic BP were 11.0 mmHg for baxdrostat and 9.6 mmHg for lorundrostat. A significant suppression of cortisol production was found for LCI699 (osilodrostat) but not for baxdrostat, lorundrostat, BI 690517 (vicadrostat) or dexfadrostat. Studies on BI 690517 showed a reduction in urine–albumin–creatinine ratio, indicating renal protection. ASIs may increase potassium levels. We conclude that ASIs have promising BP-lowering effects with very limited effects on cortisol production and offer reno-protective effects in chronic kidney disease. Studies on hypertensive target organ damage and cardiovascular outcomes are, however, lacking.

1 | Introduction

1.1 | Background

High systolic blood pressure (BP) is globally a major risk factor for mortality [1]. The age-standardized prevalence of hypertension (systolic BP ≥ 140 mmHg or diastolic BP ≥ 90 mmHg) is 33%.

Because of population growth and an increasing proportion of elderly individuals, the number of people with hypertension has increased from 650 million in 1990 to 1.3 billion in 2019 [2]. Lowering systolic BP by 10 mmHg significantly reduces the risk of major cardiovascular disease events such as coronary heart disease, stroke and heart failure, resulting in a 13% reduction in all-cause mortality [3].

Abbreviations: ABPM, ambulatory blood pressure monitoring; ACE, angiotensin-converting enzyme; ACTH, adrenocorticotrophic hormone; AE, adverse event; ARB, angiotensin II receptor antagonist; ARR, aldosterone–renin ratio; ASI, aldosterone synthase inhibitor; BID, twice a day; BMI, body mass index; BP, blood pressure; CI, confidence interval; CKD, chronic kidney disease; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; GLP, glucagon-like peptide; MR, mineralocorticoid receptor; MSDBP, mean seated diastolic blood pressure; MSSBP, mean seated systolic blood pressure; NR, not reported; QD, once a day; RAAS, renin–angiotensin–aldosterone system; RCT, randomized controlled trials; SBP, systolic blood pressure; SDBP, seated diastolic blood pressure; SE, standard error; SGLT2, sodium–glucose cotransporter-2; SSBP, seated systolic blood pressure; uACR, urine albumin–creatinine ratio.

Anders Almskou Rasmussen and Ketil Lehm Nordestgaard contributed equally to the paper.

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Summary

Poorly controlled hypertension results in cardiovascular diseases. Excess production of the hormone aldosterone worsens hypertension and organ damage. Aldosterone synthase inhibitors (ASIs) constitute a novel group of medications, and five different ASIs have now been tested in clinical trials. This article systematically assesses the effects of ASIs on blood pressure, and in patients with difficult-to-treat hypertension, ASIs can reduce systolic blood pressure by 10 mmHg. Moreover, ASIs may protect kidney function by decreasing protein excretion. In terms of adverse events, ASIs can increase plasma potassium, whereas one ASI was found to decrease cortisol production. ASIs may improve future hypertension treatment.

Unfortunately, only 23% of women and 18% of men with hypertension gain sufficient BP control. This is due mainly to underdiagnosis; however, even among those diagnosed with hypertension, 61% of women and 63% of men are either untreated or receive treatment without achieving BP control [4]. Among the treated patients, 5%–10% suffer from apparent resistant hypertension, defined as uncontrolled BP remaining at ≥ 130 -mmHg systolic BP and/or ≥ 80 -mmHg diastolic BP despite concurrent prescription of three or four antihypertensive drugs of different classes or controlled BP with the prescription of ≥ 4 antihypertensive drugs, including a thiazide diuretic and all medications at maximally tolerated doses [5, 6]. This suggests that the currently available antihypertensive agents are insufficient for the treatment of certain hypertensive patients.

The primary objective of this systematic review is to update the current knowledge about the BP-lowering effects of aldosterone synthase inhibitors (ASIs) in humans. The secondary objective is to assess the effects of ASIs on cortisol production and their renal protective properties.

1.2 | Aldosterone

The renin–angiotensin–aldosterone system (RAAS) plays a central role in BP regulation because of angiotensin II's effect on vascular tone and aldosterone's effect on electrolyte and water homeostasis. Primary aldosteronism is a condition with uncontrolled high aldosterone production and, consequently, suppressed renin. However, excess aldosterone is also seen in other patients, including some with treatment-resistant hypertension, without meeting the diagnostic criteria for primary aldosteronism [7, 8]. Excess aldosterone can lead to the worsening of hypertension and contribute to hypertensive organ damage in terms of fibrosis and dysfunction of the myocardium, vasculature and kidneys [9]. Therefore, treatment of excess aldosterone is not only of interest in hypertension but also in patients with chronic kidney disease (CKD) or heart failure [10, 11].

Figure 1 illustrates the RAAS system and the contribution of aldosterone to BP regulation. Renin, produced in the kidneys, converts angiotensinogen from the liver to angiotensin I. This intermediate is converted to angiotensin II by angiotensin-converting enzyme (ACE). Angiotensin II is the main stimulant of the adrenal production of aldosterone, which is catalysed by aldosterone synthase (CYP11B2) [11]. In the kidneys, aldosterone binds to the mineralocorticoid receptor (MR). This leads

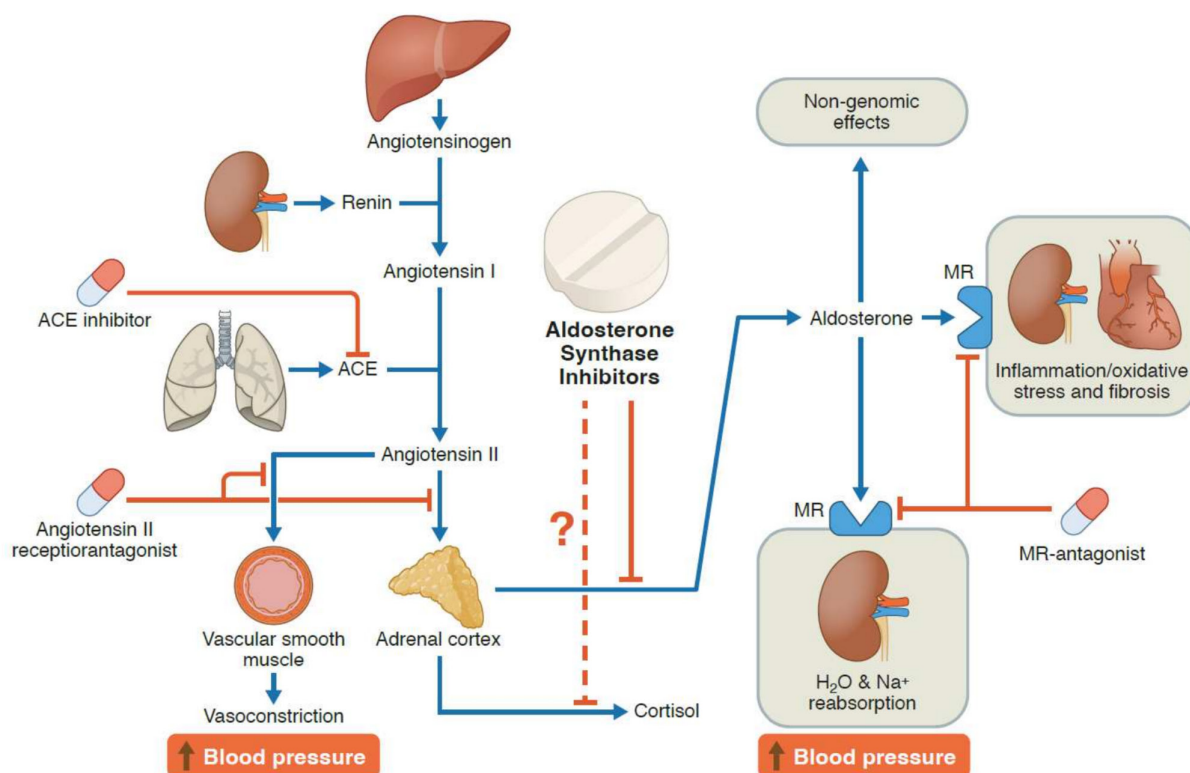


FIGURE 1 | Illustration of the RAAS, the pharmacological agents targeting RAAS including aldosterone synthase inhibitors and the downstream effects of aldosterone.

to tubular uptake of sodium (Na⁺) and excretion of potassium (K⁺) and hydrogen (H⁺). Sodium is the primary determinant of extracellular fluid volume, and water is passively absorbed as sodium is reabsorbed, leading to increased BP [12]. However, aldosterone affects multiple other cells expressing MR such as cardiomyocytes, cardiac fibroblasts and vascular smooth muscle cells [13]. Furthermore, aldosterone triggers nongenomic effects that occur independently of MR activation. It has been suggested that G protein-coupled oestrogen receptor-1 (GPER-1) may mediate some of the rapid effects of aldosterone [14], but the MR-independent effects are not fully understood and may be dependent on the degree of vascular oxidative stress. In unstressed vasculature with normal endothelial function, aldosterone causes vasodilation, whereas in vasculature with inflammation, atherosclerosis and high oxidative stress, aldosterone causes vasoconstriction and increased oxidative damage [15]. In another study, aldosterone at high concentrations, as observed in patients with heart failure, was shown to induce a negative inotropic effect. The effects on vascular function were similar, with aldosterone enhancing the vasoconstrictive response to angiotensin II in isolated human coronary arteries and porcine renal arteries. Notably, these effects were not blocked by MR antagonists [16]. In patients with risk factors such as hypertension, heart failure, high oxygen tension and high salt intake, excess aldosterone might induce rapid vasoconstriction and an increase in vascular stiffness, contributing to elevated BP [15, 17, 18].

In current antihypertensive treatment, the RAAS is directly targeted by ACE inhibitors, angiotensin II receptor blockers (ARBs), and MR antagonists. However, even long-term treatment with ACE inhibitors and ARB do not sufficiently inhibit aldosterone synthesis [8, 19], and as mentioned above, MR antagonist only inhibits the effects of aldosterone mediated through the MR and not the MR-independent nongenomic effects.

The ASIs constitute a novel therapeutic group that may overcome some of these problems. ASIs specifically block CYP11B2, the enzyme catalysing the final step in the synthesis of aldosterone and therefore reducing plasma aldosterone concentrations by 50%–70% [20], as opposed to MR antagonists, which increase circulating aldosterone levels. This new principle of RAAS blockade could therefore improve current antihypertensive treatment regimens. A recent meta-analysis by Marzano et al. based on 1,440 patients with primary hypertension or CKD from seven published RCTs demonstrated an overall reduction in office systolic BP of 6.3 mmHg, which was dependent on dose [21]. Our systematic review also reports data from trials in patients with primary hyperaldosteronism and trials in patients with diabetic kidney disease in addition to data from the HALO trial [22].

1.3 | Cortisol Synthesis

Aldosterone synthase (CYP11B2) shares 93% homology with cortisol synthase (CYP11B1) (14). The first generation of ASIs had a low specificity towards CYP11B2; consequently, it also suppressed the production of cortisol. These are now being investigated for their potential to decrease cortisol levels in

patients with Cushing's disease [23]. The impact of ASIs on cortisol levels is very important as cortisol suppression is unwanted and can cause major adverse effects. However, cortisol levels vary throughout the day, and serum cortisol is consequently an inadequate marker for cortisol synthase inhibition. The adrenocorticotrophic hormone (ACTH) stimulation test can better investigate the ability of the adrenal glands to produce cortisol. In this test, the plasma cortisol levels are measured before and after an injection of ACTH and are used to examine if cortisol synthesis is suppressed [24].

1.4 | Kidney Protection

Hypertension is both a risk factor for and consequence of CKD. Up to 90% of CKD patients have hypertension, which contributes to the progression of kidney disease [6]. The prevalence of apparent resistant hypertension is 23% among CKD patients and 56% among renal transplant patients, underscoring the challenge of treating hypertension in these patient categories [6]. Reduction in albuminuria has been linked to a decrease in the risk of end-stage kidney disease across various cohorts and is therefore often used as a surrogate marker of a drug's renoprotective properties [25].

ACE inhibitors and ARBs are first-line antihypertensive agents in patients with CKD and albuminuria, as they reduce proteinuria partly independent of their BP-lowering effects [26]. Addition of sodium–glucose cotransport Type 2 (SGLT2) inhibitors and the non-steroid MR antagonist finerenone can further reduce the risk of CKD progression [27]. Having the complex effects of aldosterone in mind, the inhibition of aldosterone synthesis could also be beneficial in CKD as an alternative to MR antagonists.

Inhibition of aldosterone synthesis offers a promising therapeutic approach by reducing both MR-dependent and -independent effects of aldosterone. Although the MR-independent mechanisms are not fully understood, they likely involve rapid nongenomic actions that contribute to vascular dysfunction, oxidative stress and increased vasoconstriction, possibly resulting in hypertension. Aldosterone synthase inhibition may improve treatment outcomes beyond those achievable with current RAAS-targeted therapies, including MR antagonists, particularly in patients with resistant hypertension or CKD.

2 | Methods

This review is registered with PROSPERO (registration number: CRD42024586848) and conducted in accordance with PRISMA guidelines to ensure transparency and reproducibility.

2.1 | Criteria for Inclusion of Studies (PICO)

The following PICO criteria were applied to find eligible studies:

- Population: Human adults ≥ 18 years of age.
- Intervention: Any ASI.

- Comparator: Placebo, baseline or another antihypertensive agent.
- Outcome: Change in systolic BP or/and diastolic BP.

2.2 | Search Strategy and Eligibility Criteria

On the 30th of August 2024, we searched PubMed, Embase, [Clinicaltrials.gov](https://clinicaltrials.gov) and clinicaltrials.eu to find relevant studies for inclusion. On the 15th of January 2025, we updated the search with studies published between 30th of August 2024 and 15th of January 2025. Randomized controlled trials (RCTs) and clinical trials meeting PICO were eligible for inclusion. Studies with only abstracts available or clinical trials with only results were included if relevant data were presented. Studies with overlapping populations were included only if they reported results on systolic or diastolic BP changes in a population distinct from the original study. The search was conducted with the following terms in all fields and with no filters used: ‘((Aldosterone synthase inhibitor) OR (Baxdrostat) OR (Lorundrostat) OR (BI 690517) OR (Dexfadrostat) OR (LCI699)) AND ((Hypertension) OR (Blood pressure))’. Any ASI found in the initial literature search tested in humans has been included independently in the search string.

2.3 | Study Selection

The two reviewers (A.A.R. and K.L.N.) independently selected the studies using the systematic review tool Covidence (www.covidence.org). Duplications were identified by an automation tool in Covidence and manually compared to exclude duplicates. Further duplications, which Covidence did not identify, were found and excluded manually. The machine learning model of Covidence identified apparent non-RCT studies that were excluded after a manual title/abstract screening. The remaining studies were screened according to title and abstract. Studies were excluded if it was possible based on title and abstract to determine if they were not a clinical trial, RCT or not meeting PICO criteria. Lastly, studies were excluded based on full text assessment if they did not meet PICO criteria. Any discrepancies were resolved by agreement.

2.4 | General Characteristics

Key study characteristics, including sample size, population characteristics, outcome measures and treatment duration, were extracted by one reviewer (A.A.R.) and subsequently cross-checked for accuracy by the second reviewer (K.L.N.). The number of patients presented under general characteristics is the number at randomization. The number of patients presented in the results table is the number of patients treated with the given dose.

2.5 | Primary Outcome

The main result is a change in BP. In the included studies, all but one present BP data as changes in seated systolic BP and seated diastolic BP. Two of the included studies present results

regarding changes in 24-h ambulatory BP measurements. To ensure comparable data, we choose to extract seated/consultation/ambulatory BP in mmHg. If not available, data on 24-h ambulatory BP in mmHg were extracted. Any variation of a seated BP measurement was accepted and referred to as seated BP without further specification.

2.6 | Comparator

In the data extraction, our priority was to extract baseline placebo-corrected changes in systolic and diastolic BP. If placebo-corrected results were not available, the BP changes of both the intervention and placebo groups were extracted. If no results of the placebo group were available, only the results of the intervention group were extracted.

2.7 | Dose of Intervention

We aim to compare the effects of different ASIs; therefore, the doses are not directly comparable. The available literature does not give sufficient data on the exact dose–response relations to determine a dose corresponding to each ASI’s maximum effect. The dose extracted from the studies corresponds to the largest reduction in systolic BP. This dose has been used for data extraction for systolic and diastolic BP changes.

2.8 | Secondary Outcomes

2.8.1 | Cortisol Synthesis

To investigate the effect of ASIs on cortisol synthesis, data have been extracted, with results from ACTH stimulation tests as the first priority and the change from baseline in serum or urinary cortisol as the secondary priority. The results are presented as a significant or non-significant change in cortisol measures.

2.8.2 | Hyperkalaemia

Data on the prevalence of hyperkalaemia were extracted from the included studies to assess the risk of hyperkalaemia associated with ASI treatment. The prevalence of hyperkalaemia in a combined dosage group was compared with the prevalence observed in the respective placebo group. Any reported definition or threshold value for hyperkalaemia was accepted and displayed alongside the results. Furthermore, treatment withdrawal due to hyperkalaemia was used as an indicator for severity, as this outcome was most consistently reported across the included studies.

2.8.3 | Renal Protection

To investigate the reno-protective properties of ASIs, we extracted results on kidney damage parameters. Our priority was results on albuminuria or proteinuria, but all kidney parameters

have been accepted. The authors have presented the most important results of the studies.

2.8.4 | Ongoing Trials

Ongoing trials were identified on [ClinicalTrials.gov](https://clinicaltrials.gov) and [ClinicalTrials.eu](https://clinicaltrials.eu) during our primary literature search, employing the same search strategy and PICO criteria. Ongoing trials that met inclusion criteria but did not present results have been presented in Table S1, with information on title, ASI, comparator, population, study type, number of patients, BP outcome and estimated completion date. One reviewer (K.L.N.) extracted these data and then cross-checked them for accuracy by a second reviewer (A.A.R.).

3 | Results

The initial literature search identified 548 records, and the updated literature search added 33 records to a total of 581 records. After exclusion of 122 duplications, the Covidence automation tool identified 247 ineligible records, which were all excluded after independent title/abstract screening by the two reviewers. We screened the remaining 212 records and excluded 136 based on title and abstract. The full text was assessed for 76 articles, leading to the exclusion of 64 studies and the inclusion of 12 studies. Since our primary literature search, one of the ongoing trials has been published [28] and therefore subsequently included, resulting in a total of 13 studies in the review (Figure 2).

Table 1 presents the general characteristics of the included studies. Table 2 presents the ASI's BP-lowering properties in each of the included studies, along with the corresponding comparator, dose and number of patients. The secondary outcomes of the review regarding effects on cortisol synthesis, effects on plasma potassium and renal protective properties are presented in Table 3 and Figure 4 and the results section, respectively.

3.1 | General BP-Lowering Effects of ASIs

The included studies vary considerably regarding study population, intervention, dose and comparator, but common to all studies is a reduction in systolic and diastolic BP in patients treated with ASI. The results of BP reductions are presented in Table 2 and illustrated as a bar chart in Figure 3. Seven of the included studies show a ≥ 10 -mmHg reduction in systolic BP, and the remaining six studies show a 4- to 10-mmHg reduction. Likewise, seven of the included studies showed a reduction of ≥ 5 mmHg in diastolic BP, and four showed a reduction in diastolic BP of 0.9–5 mmHg. Two studies did not present data on changes in diastolic BP [28, 36].

3.2 | BP-Lowering Effects of LCI699

Five of the included studies provide results of BP reduction with LCI699 (also known as osilodrostat). All studies show a

decrease in both systolic and diastolic BP, but only the study by Andersen et al. demonstrates a significant placebo-corrected effect, with reductions in systolic BP of 12.5 mmHg ($p=0.022$) and diastolic BP of 9.2 mmHg ($p=0.007$) [30]. Calhoun et al. investigated the effect of LCI699 in patients with primary hypertension and presented a nonplacebo-corrected significant reduction in systolic BP of 12.6 mmHg and diastolic BP of 7.1 mmHg [29], whereas in the placebo group, systolic and diastolic BP both decreased by 2.6 mmHg. The investigation by Karns et al. demonstrated a placebo-corrected non-significant reduction of systolic and diastolic BP among patients with resistant hypertension [31]. The two trials without placebo groups investigated the effect of LCI699 in patients with primary aldosteronism [32] and Cushing's disease [33], respectively. Both studies were able to demonstrate the BP-lowering effects of LCI699.

3.3 | BP-Lowering Effects of Dexfadrostat

One study investigating the effect of dexfadrostat has been included [34]. In this Phase II trial, Wuerzner et al. studied patients with primary aldosteronism and hypertension. Results were only available regarding seated systolic BP as a combined group receiving treatment with either 4, 8 or 12 mg once daily. The study did not include a placebo group. The combined dosage group showed a significant reduction compared with baseline in both systolic and diastolic BP: 10.7 mmHg (95% CI [7.9, 13.6]) and 5.7 mmHg (95% CI [3.5, 7.8]), respectively.

3.4 | BP-Lowering Effects of Lorundrostat

Two studies included on lorundrostat investigated the effect in the same population of patients with resistant hypertension. Laffin et al. found a significant placebo-corrected reduction in systolic and diastolic BP of 9.6 mmHg (90% CI [3.4, 15.8]) and 5.5 mmHg (90% CI [1.5, 9.4]), respectively [35]. In a substudy, Rodman et al. reported on patients with BMI > 30 kg/m² [36]. In this subgroup, a significant placebo-corrected reduction in systolic BP of 16.7 mmHg (90% CI [7.9, 25.5]) was found, whereas no data on diastolic BP were presented in the abstract. More recently, Laffin et al. reported a significant placebo-adjusted reduction in systolic BP of 7.9 mmHg (97.5% CI [13.3, 2.6]) in patients with hypertension despite treatment with two to five antihypertensive agents. Diastolic BP outcomes were not reported in this study.

3.5 | BP-Lowering Effects of Baxdrostat

Two studies on baxdrostat examined the BP-lowering effects of a 2 mg daily dose in patients with resistant hypertension. Freeman et al. found placebo-corrected significant reductions in systolic and diastolic BP of 11.0 (95% CI [5.5, 16.4]) and 5.2 mmHg (95% CI [1.6, 8.8]), respectively [37]. The HALO study described nonplacebo-corrected reductions in systolic and diastolic BP of 19.8 and 5.4 mmHg, respectively. However, the HALO study also noticed large BP reductions in the placebo group of 16.6 mmHg systolic and 5.9 mmHg diastolic [22].

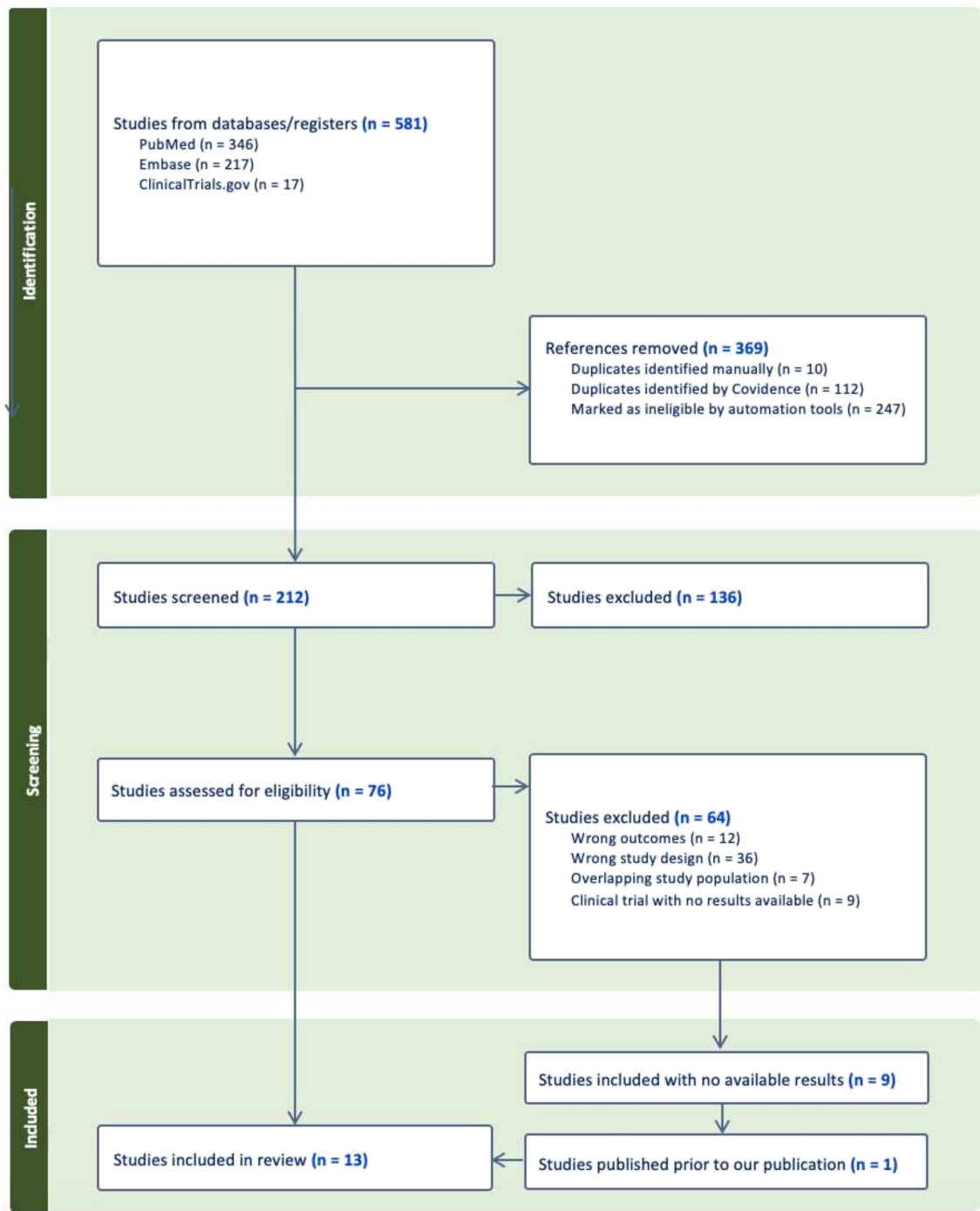


FIGURE 2 | PRISMA flowchart of the study inclusion process.

3.6 | BP-Lowering Effects of BI 690517

Two of the included studies investigated the effects of BI 690517 (also known as vicadrostat) in patients with CKD. Tuttle et al. found non-significant placebo-corrected changes in systolic and diastolic BP of -6.03 mmHg (95% CI $[-12.44, 0.38]$) and -0.94 mmHg (95% CI $[-4.31, 2.43]$), respectively, using a daily dose of 20 mg [39]. Bornstein et al. found a significant, but not placebo-corrected, systolic BP change of -6.38 mmHg (95% CI $[-11.64, -1.11]$) and a non-significant diastolic BP change of -2.95 mmHg (95% CI $[-6.33, 0.43]$) among patients with CKD and Type 2 diabetes when receiving a daily dose of 3 mg [38].

3.7 | Effects of Aldosterone Synthase Inhibition on Cortisol

The effects of ASI treatment on cortisol production are presented in Table 3. Three of the included studies on LCI699 showed a significant impact of this drug on cortisol synthesis as measured by the ACTH stimulation test or urinary-free cortisol. However, none of the studies on baxdrostat, lorundrostat, BI 690517 or dexfadrostat could detect any influence on cortisol levels measured by the ACTH stimulation test, changes in serum cortisol or 24-h urinary cortisol. Four of the included studies did not present results of any cortisol measure.

TABLE 1 | General characteristics of the included studies.

Author	ASI	Study type	Randomized patients (n)	Population characteristics	Outcome measures	Treatment period
Calhoun et al. 2011 [29]	LCI699	Multicentre, double-blind Phase II RCT	524	Stage 1 + 2 hypertension, either untreated or treated with ≤ 2 antihypertensive agents	Primary: Change from baseline in clinic BPs (placebo corrected) Secondary: Effect on aldosterone, renin and cortisol. ACTH stimulation test and safety and tolerability	8 weeks
Andersen et al. 2012 [30]	LCI699	Multicentre, double-blind Phase II RCT	63	Primary hypertension treated with minimum of 1 antihypertensive agent	Primary: Maximal tolerated dose with respect to cortisol depression measured by ACTH stimulation test Secondary: Pharmacokinetics, safety and tolerability, proportion of patients achieving BP control, MSSBP and MSDBP change	6 weeks
Karns et al. 2013 [31]	LCI699	Multicentre, double-blind RCT study	155	Resistant hypertension	Primary: Change in MSSBP compared with placebo Secondary: MSDBP change, 24-h BP, correlation between BP reduction and changes in RAAS biomarkers	8 weeks
Amar et al. 2013 [32]	LCI699	Single-centre, single-blind, placebo-controlled Phase IIa clinical trial	14	Hypertension and primary aldosteronism	Primary: Blood pressure, hormonal effects, tolerability and safety	4 weeks
Bertagna et al. 2014 [33]	LCI699	Multicentre, proof of concept clinical trial	12	Moderate to severe cushing's disease defined as urinary free cortisol > 1.5 times upper normal limit	Primary: The proportion of patients with urinary free cortisol $\geq$ upper normal limit or who had a $\geq 50\%$ decrease from baseline at Day 70 Secondary: Pharmacodynamic parameters of the hypothalamic-pituitary-adrenal and gonadal axes, change in SBP and DBP	10 weeks
Wuerzner et al. 2022 [34]	Dexfadrostad	Multicentre, double blind, randomized Phase II trial	35	Primary aldosteronism and SSBP > 145 mmHg and eGFR > 45 mL/min/1.73 m ²	Primary: Change in ARR and change in mean 24-h ambulatory SBP for all dose groups Secondary: Change in SSBP and plasma ARR in each of the dose groups	8 weeks

(Continues)

TABLE 1 | (Continued)

Author	ASI	Study type	Randomized patients (n)	Population characteristics	Outcome measures	Treatment period
Laffin et al. 2023 [35]	Lorundrostat	Multicentre, double blind, dose-ranging RCT	200	Uncontrolled hypertension while taking ≥ 2 antihypertensive agents	Primary: change in SSBP from baseline Secondary: Change in SDBP, 24-h ambulatory BP and proportion of patients achieving BP control	8 weeks
Rodman et al. 2023 ^a [36]	Lorundrostat	Substudy of Laffin et al.	NR	Uncontrolled hypertension taking ≥ 2 antihypertensive agents and BMI ≥ 30	Primary: Change in SSBP among patients with BMI ≥ 30	8 weeks
Halvorsen 2023 [22] (HALO) ^b	Baxdrostat	Multicentre, double-blind Phase II RCT	249	Patients taking 1 or 2 antihypertensive agents without BP control	Primary: Change from baseline in mean SSBP Secondary: Change from baseline in mean SDBP, change in 24-h serum and urine renin and aldosterone	8 weeks
Freeman et al. 2023 [37]	Baxdrostat	Multicentre, double-blind Phase II RCT	275	Patients treated with ≥ 3 antihypertensive agents (at least 1 diuretic) without BP control	Primary: Change in SSBP with baseline (placebo corrected) in each dose group Secondary: Change in SDBP with baseline (placebo corrected) in each dose group Percentage of patients with BP control at the end of treatment	12 weeks
Bornstein et al. 2024 [38]	BI 690517	Multicentre, double-blind Phase I RCT	62	Type 1 or 2 diabetes and CKD treated with both insulin, GLP1 analogue, or oral antidiabetic agent and ACE inhibitor or ARB	Primary: Safety measured as proportion patients with drug-related AEs Secondary: Change from baseline in UACR, exploratory biomarkers (BP), pharmacokinetic biomarkers	4 weeks
Tuttle et al. 2024 [39]	BI 690517	Multicentre, double blind, Phase II RCT	586	CKD with or without Type 2 diabetes	Primary: Change from baseline in UACR in the first morning void Secondary: Proportion of patients with either > 15% or > 30% decrease in uACR from baseline	14 weeks
Laffin et al. 2025 [28]	Lorundrostat	Multicentre, prospective, double-blind RCT	285	Patients on stable treatment with 2–5 antihypertensives and elevated office BP	Primary: Change in 24-h average systolic blood pressure from baseline at Week 12 Secondary: Change in 24-h average systolic blood pressure from baseline to Week 4	12 weeks

Abbreviations: ACE—angiotensin-converting enzyme; ACTH—adrenocorticotrophic hormone; AE—adverse event; ARB—angiotensin II receptor blocker; ARR—aldosterone–renin ratio; ASI—aldosterone synthase inhibitor; BID—twice a day; BMI—body mass index; BP—blood pressure; CKD—chronic kidney disease; DBP—diastolic blood pressure; eGFR—estimated glomerular filtration rate; GLP—glucagon-like peptide; MSDBP—mean seated diastolic blood pressure; MSSBP—mean seated systolic blood pressure; NR—not reported; QD—once a day; RAAS—renin–angiotensin–aldosterone system; RCT—randomized controlled trial; SBP—systolic blood pressure; SSBP—seated systolic blood pressure; uACR—urine–albumin–creatinine ratio.

^aOnly poster and/or abstract available.

^bOnly results from clinical trial available.

TABLE 2 | Blood pressure-lowering effects of aldosterone synthase inhibitors.

Author	Specific population	Intervention	Comparator	Daily dose	Number of patients (placebo)	Change in SSBP [95% CI], <i>p</i> compared with placebo or SE	Change in SDBP (90/95% CI), <i>p</i> compared with placebo or SE
Calhoun et al. 2011 [29]	Primary hypertension	LCI699	Baseline (not placebo corrected)	1 mg QD	77 (67)	−12.6 mmHg, <i>p</i> = 0.0001 (compared with placebo)	−7.1 mmHg, <i>p</i> = 0.0012 (compared with placebo)
		Placebo	Baseline	0 mg	67	−2.6 mmHg	−2.6 mmHg
Andersen et al. 2012 [30]	Primary hypertension	LCI699	Baseline (placebo corrected)	2 mg (1 mg BID)	13 (13)	−12.5 mmHg, <i>p</i> = 0.022	−9.2 mmHg, <i>p</i> = 0.007
Karns et al. 2013 [31]	Resistant hypertension	LCI699	Baseline (placebo corrected)	1 mg QD	26 (33)	−4.3 mmHg, <i>p</i> > 0.1	−1.2 mmHg, <i>p</i> > 0.1
Amar et al. 2013 [32]	Primary aldosteronism	LCI699	Baseline (not placebo corrected)	2 mg (1 mg BID)	14 (no placebo)	−4.1 mmHg ^c , 95% CI [−8.1, −0.1]	−2.1 mmHg ^c , 95% CI [−4.5, 0.3]
Bertagna et al. 2013 [33]	Cushing disease	LCI699	Baseline (not placebo corrected)	100 mg (50 mg BID)	12 (no placebo)	−10.0 mmHg, SE = 4.3	−6.0 mmHg, SE = 4.3
Wuerzner et al. 2022 [34]	Primary aldosteronism	Dexfadrost	Baseline (not placebo corrected)	4, 8 or 12 mg QD combined	35 (no placebo)	−10.7 mmHg, 95% CI [−13.6, −7.9]	−5.7 mmHg, 95% CI [−7.8, −3.5]
Laffin et al. 2023 [35]	Resistant hypertension	Lorundrost	Baseline (placebo corrected)	50 mg QD	28 (29)	−9.6 mmHg, 90% CI [−15.8, −3.4]	−5.5 mmHg, 90% CI [−9.4, −1.5]
Rodman et al. 2023 ^a [36]	Resistant hypertension and BMI > 30	Lorundrost	Baseline (placebo corrected)	50 mg QD	15 (NR)	−16.7 mmHg, 90% CI [−25.5, −7.9]	NR
Halvorsen 2023 (HALO) ^b [22]	Resistant hypertension	Baxdrostat	Baseline (not placebo corrected)	2 mg QD	55 (61)	−19.8 mmHg, SE = 1.67	−5.4 mmHg, SE = 1.23
		Placebo	Baseline	0 mg	64	−16.6 mmHg, SE = 1.58	−5.9 mmHg, SE = 1.17
Freeman et al. 2023 [37]	Resistant hypertension	Baxdrostat	Baseline (placebo corrected)	2 mg QD	67 (69)	−11.0 mmHg, 95% CI [−16.4, −5.5]	−5.2 mmHg, 95% CI [−8.7, −1.6]
Bornstein et al. 2024 [38]	Diabetes and CKD	BI 690517	Baseline (not placebo corrected)	3 mg QD	18 (8)	−6.38 mmHg, 95% CI [−11.64, −1.11]	−2.95 mmHg, 95% CI [−6.33, 0.43]
		Placebo	Baseline	0 mg	8	−1.89 mmHg, 95% CI [−9.82, 6.05]	0.16 mmHg, 95% CI [−4.83, 5.15]
Tuttle et al. 2024 [39]	CKD	BI 690517	Baseline (placebo corrected)	20 mg QD	54 (57)	−6.03 mmHg, 95% CI [−12.44, 0.38]	−0.94 mmHg, 95% CI [−4.31, 2.43]

(Continues)

TABLE 2 | (Continued)

Author	Specific population	Intervention	Comparator	Daily dose	Number of patients (placebo)	Change in SSBP [95% CI], <i>p</i> compared with placebo or SE	Change in SDBP (90/95% CI), <i>p</i> compared with placebo or SE
Laffin et al. 2025 [28]	Elevated office BP on treatment with 2–5 antihypertensives	Lorundrostat	Baseline (placebo corrected)	50 mg daily	95 (94)	–7.9 mmHg, 97.5% CI [–13.3, –2.6]	NR

Abbreviations: BID—twice a day; CI—confidence interval; CKD—chronic kidney disease; NR—not reported; QD—once a day; SDBP—seated diastolic blood pressure; SE—standard error; SSBP—seated systolic blood pressure.

^aOnly poster and/or abstract available.

^bOnly results from clinical trial available.

^c24-h ambulatory blood pressure.

3.8 | Prevalence of Hyperkalaemia in Patients Treated With ASI

Figure 4 presents the prevalence of hyperkalaemia in the combined dosage groups across the included studies. The prevalence ranges from 0% to 13.5%, with the highest prevalence reported by Laffin et al. in patients with resistant hypertension treated with lorundrostat [35]. The most recent study on lorundrostat by Laffin et al. reported a prevalence of 10.1% [28]. The prevalence of hyperkalaemia was 3.5% in both studies investigating baxdrostat [22, 37]. In studies investigating the effects of BI 690517 in patients with CKD, Bornstein et al. and Tuttle et al. reported hyperkalaemia prevalences of 4.4% and 13.3%, respectively [38, 39]. In the study by Tuttle et al., the prevalence of hyperkalaemia in the placebo group was 6.8%, the highest observed among a placebo group in any of the included studies. Less hypercalcemia was observed in studies on LCI699 in patients with primary hypertension, where Calhoun et al. and Andersen et al. reported prevalence rates of 3.1% and 2.0%, respectively [29, 30].

Withdrawal due to hyperkalaemia was rare but inconsistently reported across studies. No withdrawals occurred in the studies by Calhoun et al. or Wuerzner et al., whereas single cases were noted by Andersen et al. and Bornstein et al. Freeman et al. reported three temporary discontinuations, and Tuttle et al. reported 4% discontinuation due to hyperkalaemia. The most recent study by Laffin et al. did not report withdrawals but stratified severity, with 1% having serum potassium of > 6.5 mmol/L.

3.9 | Kidney Protection by Aldosterone Synthase Inhibition

Two of the included studies present results on kidney damage parameters. Tuttle et al. studied the effects of BI 690517 in CKD, and patients with a reduction in urine albumin–creatinine ratio (uACR) of more than 30% from baseline were defined as responders [39]. The study found a placebo-corrected odds ratio of being a responder of 6.09 (95% CI [2.64, 14.08]) with a 10-mg daily dose of BI 690617. Bornstein et al. investigated the effects of BI 690517 in patients with concomitant CKD and diabetes (diabetic kidney disease) [38]. A 40-mg dose of BI 690517 caused a ≥20% decrease from baseline in uACR in 80% of patients, whereas placebo treatment only showed the same effect in 37.5%. None of the other included studies present data on renal function parameters.

In the study by Laffin et al., a reduction in eGFR of 13%–15% was observed among patients treated with lorundrostat, compared with a 3% decline in the placebo group. Following a washout period, eGFR increased in the lorundrostat group, while remaining stable in the placebo group. This indicates the well-known effects of aldosterone inhibition on glomerular haemodynamics, which are also observed with MR antagonists.

4 | Discussion

This systematic literature review demonstrates promising BP-lowering effects of several ASIs across various patient categories affected by high BP. The most pronounced

TABLE 3 | The effect of aldosterone synthase inhibitors on cortisol synthesis.

Author	ASI	ACTH stimulation test	Other cortisol measure
Calhoun et al. 2011 [29]	LCI699	Significant cortisol depression	NR
Andersen et al. 2012 [30]	LCI699	Significant cortisol depression	NR
Karns et al. 2013 [31]	LCI699	Not performed	NR
Amar et al. 2013 [32]	LCI699	Not performed	NR
Bertagna et al. 2013 [33]	LCI699	Not performed	Significant normalization or reduction > 50% in urinary free cortisol
Wuerzner et al. 2022 [34]	Dexfadrostat	Not performed	No significant change in serum cortisol compared with baseline
Laffin et al. 2023 [35]	Lorundrostat	Normal cortisol response	No significant change in serum cortisol compared with baseline
Rodman et al. 2023 ^a [36]	Lorundrostat	Not performed	NR
Halvorsen 2023 (HALO) ^b [22]	Baxdrostat	Not performed	NR
Freeman et al. 2023 [37]	Baxdrostat	Not performed	No significant change in serum cortisol compared with baseline
Bornstein et al. 2024 [38]	BI 690517	Not performed	No significant change in plasma cortisol or 24-h urinary cortisol compared with baseline
Tuttle et al. 2024 [39]	BI 690517	Not performed	No significant change in morning serum cortisol compared with baseline
Laffin et al. 2025 [28]	Lorundrostat	Not performed	No patients developed hypercortisolism, defined as serum cortisol > 35 µg/dL confirmed by 24-h urinary free cortisol

Abbreviations: ACTH—adrenocorticotropic hormone; ASI—aldosterone synthase inhibitor; NR—not reported.

^aOnly poster and/or abstract available.

^bOnly results from clinical trial available.

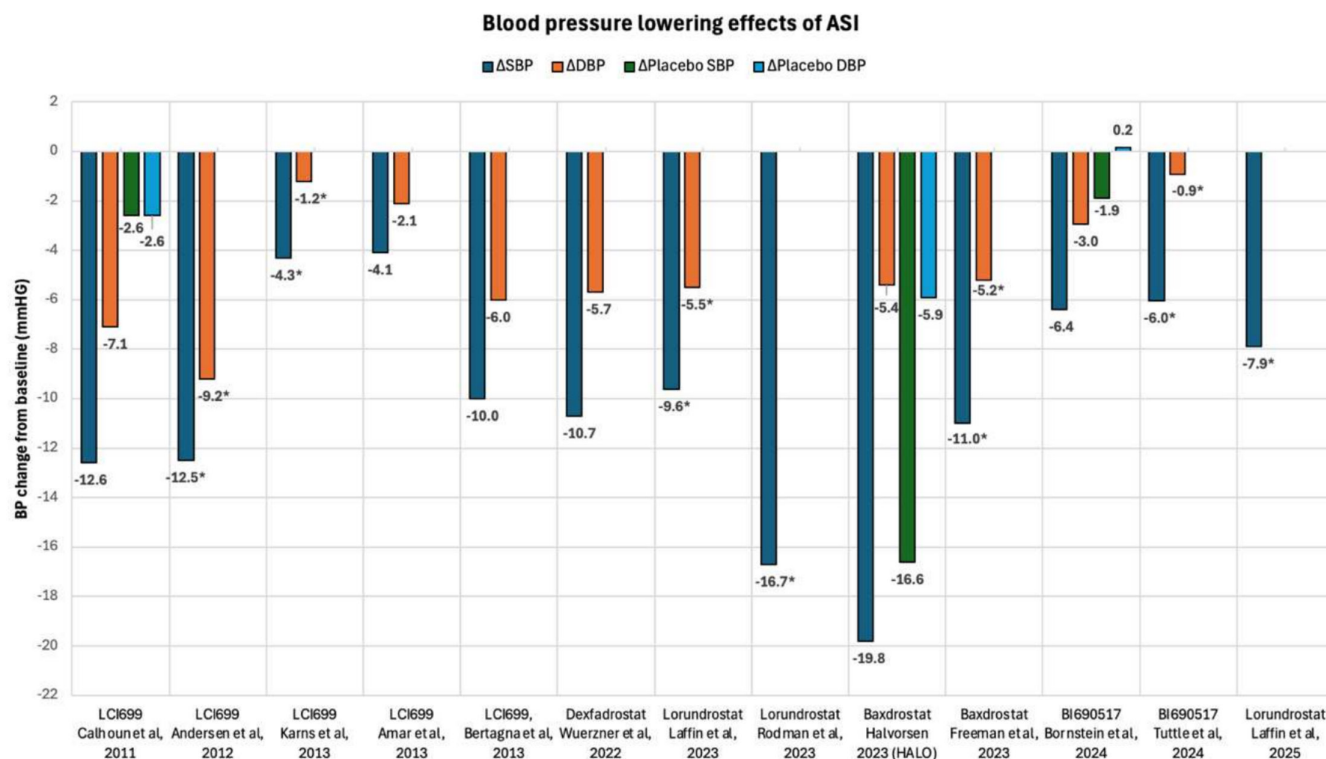


FIGURE 3 | Results from the included studies on the blood pressure-lowering effects of aldosterone synthase inhibitors.

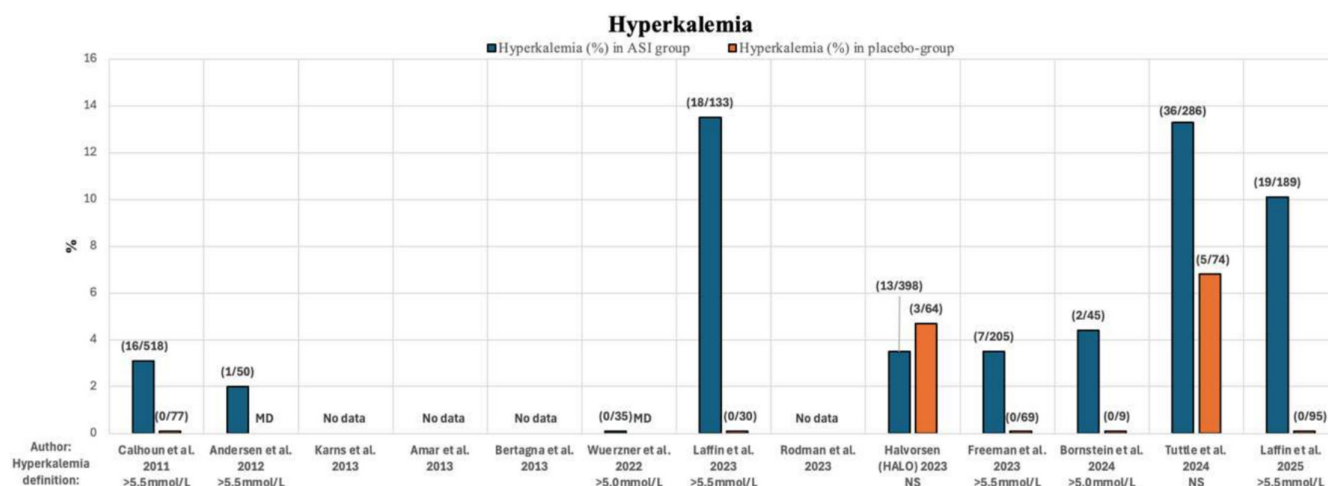


FIGURE 4 | Prevalence of hyperkalemia associated with aldosterone synthase inhibitor treatment in the included studies. “No data” indicates that the respective study did not report results on hyperkalemia. ASI—aldosterone synthase inhibitor; MD—missing data; NS—not specified.

placebo-adjusted reductions were reported in patients with apparent resistant or primary hypertension. ASIs also have effects on primary hyperaldosteronism and Cushing’s disease. In addition, ASIs have potential reno-protective properties as documented by a reduction in albumin excretion beyond the blockade with ACE inhibitors and ARBs. The new generation of ASIs shows no or minimal impact on cortisol production. These findings establish ASIs as a novel and overall safe treatment for hypertension and CKD.

4.1 | Data Extraction and Interpretation of Results

As this review attempts to present all available data on BP-lowering effects of ASIs, it also includes very diverse studies. We compare the effects of ASIs on different populations, ranging from patients with resistant or uncontrolled hypertension to patients with CKD and only mildly elevated BP or even controlled BP. Hypertensive patients have a larger potential for BP reduction, which may underestimate the BP-lowering effect of BI 690517, which was mainly studied in patients with CKD and overall well-controlled BP [38, 39]. Patients with resistant hypertension have significant challenges in achieving BP control on antihypertensive treatment and may have a smaller BP-lowering effect of ASIs compared with patients without resistant hypertension. Treatment with MR antagonists is unlikely to be a confounding factor, as all included studies either excluded patients receiving MR antagonists or implemented a washout period prior to inclusion.

The diversity of study populations highlights the broad efficacy of ASI in reducing BP across various patient groups. Moreover, it provides valuable insight into specific populations that may gain greater benefits from ASIs. An illustrative example of this phenomenon is provided by Rodman et al. who investigated a subpopulation of obese patients and reported the largest placebo-corrected reduction in systolic BP of any ASI of 16.7 mmHg [36]. This finding indicates that ASIs could be important in the future management of hypertension among obese patients.

Half of the studies included in this review do not present placebo-corrected results on change of BP. This is a major challenge,

because nonplacebo-corrected results consistently overestimate the BP-lowering effect, and many RCTs with antihypertensive agents as intervention demonstrate a BP reduction also in the placebo group. Several factors may contribute. One is a reduction in the white coat effect, where patients, over time, become familiar with the clinical setting, leading to reduced anxiety, resulting in lower BP measures. Additionally, administration of placebo treatment triggers physiological responses such as relaxation, which can contribute to reductions in BP. As a result, trials lacking placebo-controlled results may overestimate the BP-lowering effects of an ASI [40]. The most pronounced placebo effect was noticed in the HALO study, which found a reduction in systolic BP of 19.8 mmHg in the actively treated group and a reduction of 16.6 mmHg in the placebo group [22]. Therefore, nonplacebo-corrected results are more susceptible to bias and must be cautiously evaluated.

In the data extraction process, we included the dosage associated with the largest systolic BP reduction to ensure data reflecting the maximal effect of each ASI. This approach may introduce bias, as studies utilize different doses for various indications, potentially resulting in greater observed reductions in BP in some studies. For instance, data from BI 690517 included dosages of 3 mg from one study and 20 mg from another [38, 39]. On the other hand, higher dosages may not result in further BP reductions if the maximal effect is achieved at a lower dose. The inclusion of diverse dosages can, therefore, complicate the comparison of the results for each ASI and between different ASIs.

In this review, we extracted data on seated systolic BP rather than 24-h ambulatory BP recordings, even when both measurements were presented. This improves comparability between the studies, as the majority only present results on seated BP. However, office-seated BP is susceptible to white-coat hypertension [41] and may, therefore, exaggerate BP reductions by ASIs in the studies, not including placebo corrections.

We included Phase I, II and III trials to maximize data on ASI and BP-lowering. Phase I studies like the one by Bornstein et al. [38] focused on safety and pharmacokinetics with limited data

on efficacy. Phase II and III trials, like the ones by Tuttle et al. [39] and Freeman et al. [37], assessed much larger populations, providing more reliable evidence on efficacy [40].

4.2 | BP-Lowering Effects of ASIs

A previous systematic review [42] has summarized the BP-lowering effects of all available ASIs, but only among patients with resistant hypertension and only including data from the studies by Calhoun et al. [29], Freeman et al. [37] and Laffin et al. [35]. The BP-lowering effects of individual ASIs have been described in systematic reviews regarding baxdrostat and lorundrostat, and a meta-analysis has described the effects of LCI699 (now known as osilodrostat) [43–45]. The latter found a change in seated systolic BP of 8.80 mmHg and a change in seated diastolic BP of 4.94 mmHg [43] derived from data by Andersen et al. [30] and Calhoun et al. [29]. The recent meta-analysis by Marzarno et al. included data based on LCI699, BI 690515, lorundrostat and baxdrostat [21]. Our review is the first to summarize the BP-lowering effects of all available ASIs across different populations.

So far, none of the ASIs tested on humans are significantly superior to others. LCI699 has demonstrated significant BP reduction in patients with primary aldosteronism, with two studies showing notable effects, including one reporting a significant placebo-corrected reduction [29, 30]. Patients with CKD treated with BI 690517 show moderate BP reductions. However, none of these studies provide statistically significant and placebo-corrected results.

Dexfadrostat appears to have superior effects compared with LCI699 in patients with primary aldosteronism, although this is not statistically significant, and none of the included studies show results regarding the placebo-corrected BP reduction [34]. The results of our review point to large, significant placebo-corrected BP reductions in patients with resistant hypertension treated with baxdrostat and lorundrostat, whereas treatment with LCI699 only showed a small, non-significant placebo-corrected BP reduction.

According to the guidelines of the European Society of Hypertension, the MR antagonist spironolactone is recommended in the treatment of patients with resistant hypertension [46]. This recommendation is based on a placebo-corrected significant reduction in systolic BP of 8.70 mmHg in the British PATHWAY-2 study [47]. Comparable BP-lowering effects are observed for lorundrostat and baxdrostat. Thus, lorundrostat showed a placebo-corrected reduction in systolic BP of 9.6 mmHg, whereas baxdrostat demonstrated a placebo-corrected reduction of 11.0 mmHg [35, 37]. These data suggest that ASIs are at least as effective as spironolactone in resistant hypertension in terms of BP reduction. Whether spironolactone and ASIs have similar effects on hypertension-mediated organ damage remains to be established.

In contrast, the study by Karns et al. comparing LCI699 with eplerenone reported a non-significant placebo-corrected systolic BP reduction of only 4.3 mmHg with LCI699, which was considerably lower than the 9.9-mmHg reduction achieved with eplerenone in the same patient population [31]. Two additional studies

demonstrated significantly smaller systolic BP reductions with LCI699 as compared with eplerenone in patients with primary aldosteronism or primary hypertension [29, 32], further indicating an inferior BP-lowering efficacy of LCI699 compared to eplerenone [29]. No studies have compared the effects of baxdrostat, lorundrostat, dexfadrostat or BI 690517 with MR antagonists. Currently, the BP-lowering properties of ASIs and spironolactone must be considered comparable, and whether there are differences regarding effects on hypertensive organ damage remains to be established. For the moment being, no studies are testing the effects of ASIs on cardiovascular outcomes, like stroke or myocardial ischemic events, in primary or resistant hypertension.

4.3 | The Effect of ASIs on CKD

The only ASI included in this review with results on kidney function is BI 690517 (vicadrostat), which shows promising renoprotective effects in patients with CKD in terms of a reduction in urine albumin excretion. Tuttle et al. found that patients treated with BI 690517 had a significantly greater chance of reaching a $\geq 30\%$ reduction in uACR, with an odds ratio of 6.09 compared with placebo [39]. A recent post hoc analysis of the Lancet study by Tuttle et al. revealed that BI 690517 was able to reduce uACR also in patients already treated with an SGLT2i [48], demonstrating additional renoprotection to current treatment regimens. As outlined in the introduction, excess aldosterone contributes to renal fibrosis, a key factor in the progression of CKD, and ASI treatment may contribute to renal protection by inhibiting inflammatory pathways [11, 15]. Although ASIs represent a promising therapeutic approach for slowing the progression of CKD, longer-term studies with hard renal endpoints, such as reduced loss of glomerular filtration rate, are still lacking.

4.4 | The Effect of ASIs on Cortisol Synthesis

The studies of LCI699 have consistently demonstrated a depression of cortisol production, likely due to an inhibitory effect of LCI699 on cortisol synthase. This undesirable side effect probably renders LCI699 unsuitable for primary hypertension treatment, and this substance is now approved to reduce cortisol levels in Cushing's disease [49]. In contrast, the new-generation ASIs—baxdrostat, lorundrostat, dexfadrostat and BI 690517—show no signs of cortisol depression in any of the included studies. This points to their high specificity for aldosterone synthase, which is promising for their future use in hypertension treatment.

4.5 | Risk of Hyperkalaemia Associated With ASI Treatment

Hyperkalaemia is a well-documented adverse effect of RAAS inhibition [50]. Given this risk, most studies on ASIs have investigated the prevalence of hyperkalaemia. In the included studies, the prevalence of hyperkalaemia ranged from 0% to 13.5%. Most cases of hyperkalaemia were mild to moderate. Treatment discontinuation was used as a surrogate marker for the severity of hyperkalaemia. Although this is a non-specific measure, given that the threshold for discontinuation varied between studies, it was the most frequently reported and thus

served best for comparison. Treatment discontinuation due to hyperkalaemia was reported in 0% to 4% of patients across the included studies.

Direct comparisons between studies are challenging because of differences in study populations, especially regarding kidney function and treatment with different ASIs. In the studies of patients with primary aldosteronism, ASI led to the normalization of potassium levels in most patients with baseline hypokalaemia, suggesting the beneficial effect of ASIs on potassium levels in this subgroup [32, 34]. A Danish population-based study followed 262 375 new users treated with RAAS inhibitors, with 16% experiencing a first hyperkalaemia event. Thus, ASIs may carry a similar, or slightly lower, risk of hyperkalaemia compared with older RAAS-targeting drugs [51]. Further studies are necessary to evaluate the risk of hyperkalaemia associated with each ASI in different clinical settings.

5 | Limitations

Several limitations in our systematic review must be acknowledged. First, we did not systematically assess bias for the included studies. The absence of a formal bias assessment may introduce a risk of including studies with methodological weaknesses, potentially affecting the reliability of the presented results. Second, this review did not include a meta-analysis, and therefore, the results presented were not weighted based on study size, which may have overemphasized small and underemphasized large studies. This challenges the ability to draw definitive conclusions on the effect of each ASI.

Additionally, the inclusion of non-RCT studies introduces a risk of bias. Furthermore, including studies not presenting placebo-corrected results (although we reported placebo group results where available) makes comparison challenging, and in nonplacebo-corrected and non-RCT studies, the effects of ASI may be overestimated.

Finally, the available studies regarding aim, population, drug, dose and comparator are highly diverse. This limitation reduces our ability to determine which ASI achieves the largest BP reduction and evaluate clinical implications.

5.1 | Ongoing Trials

We identified eight ongoing clinical trials investigating ASIs with a change in BP as one of their outcomes. All these trials are registered on [ClinicalTrials.gov](https://www.clinicaltrials.gov). The ongoing trials will provide further insights into the BP-lowering effects of ASIs across different populations and other possible indications, such as CKD. Details of the ongoing studies are presented in Table S1.

6 | Conclusion

The ASIs constitute a novel drug class with promising antihypertensive effects across various patient categories with elevated

BP and may also be reno-protective in CKD. Both lorundrostat and baxdrostat demonstrate BP reductions comparable with spironolactone, the current standard treatment for resistant hypertension. Future studies on each ASI in relevant populations will be essential to evaluate their role in future antihypertensive and cardiorenal treatment.

Conflicts of Interest

A.A.R. and K.L.N. have no conflicts of interest. U.S. is a co-founder and Consultant Chief Scientific Officer in Initiator Pharma A/S (www.initiatorpharma.com) with salary paid in forms of stock options (no relations to the submitted work). N.H.B. is a local investigator in a study including baxdrostat (D6970C00002) supported by AstraZeneca (no relations to the submitted work).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.